Reg. No. | | | | | | | | | | |

[17BS4AE01]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE–2019
ENGINEERING MATHEMATICS FOR AEROSPACE ENGINEERS

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed.
3. Statistical tables (Normal, Students-t and Chi-square distribution) will be provided

PART-A

1. a. Prove that, for any two events A and B , **04**

$P(A - B) = P(A) - P(A \cap B)$

b. A box A contains 2 white balls and 4 black balls. Another box B contains 5 white balls and 7 black balls. A ball is transferred from box A to box B. Then a ball is drawn from box B. Find the probability that it is white. **05**

c. An office has 4 secretaries handling 20%, 60%, 15% , 5% respectively of the files of certain reports. The probabilities that they misfile such reports are respectively 0.05, 0.1, 0.1and 0.05. Find the probability that a misfiled report is caused by the first secretary. **05**

2. a. A random variable X has the following probability function for various values of x **04**

x	0	1	2	3	4	5	6	7
P(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

Find k and P(x < 6)

b. The probability that a news reader commits no mistake in reading news is $1/e^3$. Find the probability that on a particular news broad cast he commits (i) Only 2 mistakes, (ii) More than 3 mistakes (iii) At most 3 mistakes **05**

c. The marks of 1000 students in an examination follow a normal distribution with mean 70 and standard deviation 5. Find the number of students whose marks will be (i) Less than 65 (ii) More than 75 . **05**

3. a. The weights of 1500 ball bearings are normally distributed with a mean of 635gms and S.D of 1.36 gms. If 300 random samples of size 36 are drawn from this population determine the expected mean and S.D of the sampling distribution of means if sampling is done with replacement, find how many random samples would have their mean (i) between 634.76 gms and 635.24 gms (ii) greater than 635.5 gms (iii) less than 634.2 gms. **07**

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b. Ten individuals are chosen at random from a population and their heights in inches are found to be 63, 63, 66, 67, 68, 69, 70, 70, 71, 71. Test the hypothesis that the mean height of the universe is 66 inches. [$t_{0.05} = 2.262$ for 9 d.f.] **07**

4. a. The mean and standard deviation of maximum loads supported by 60 cables are 11.09 tonnes and 0.73 tonnes respectively. Find (a) 95% (b) 99% confidence limits for mean of the maximum loads of all cables produced by the company. **04**

b. A group of 10 boys fed on a diet A and another group of 8 boys fed on a different diet B for a period of 6 months recorded the following increase in weights (lbs.) **05**

Diet A	5	6	8	1	12	4	3	9	6	10
Diet B	2	3	6	8	10	1	2	8	-	-

Test whether diets A and B differ significantly regarding their effect on increase in weight.

c. 4 coins are tossed 100 times and the following results were obtained. Fit a binomial distribution for the data and test the goodness of fit ($\chi^2_{0.05} = 9.49$ for 4 d.f.) **05**

No. of heads	0	1	2	3	4
Frequency	5	29	36	25	5

PART- B

5. a. Find the unique fixed probability vector for the regular stochastic **04**

Matrix $P = \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix}$.

b. Habitual gambler is a member of two clubs A and B. He visits either of the clubs everyday for playing cards. He never visits club A on two consecutive days. But, if he visits club B on a particular day, then the next day he is as likely to visit club B or club A. Find the transition matrix of this Markov chain. Also, (i) show that the matrix is a regular stochastic matrix and find the unique fixed probability vector. (ii) If the person had visited club B on Monday, find the probability that he visits club A on Thursday. **05**

c. Let X and Y are independent random variables. X take values 2, 5, 7 with probability ½, ¼, ¼ respectively. Y take values 3, 4, 5 with the probability 1/3, 1/3, 1/3. **05**

i. Find the joint probability distribution of X and Y.

ii. Show that the covariance of X and Y is equal to zero.

6. a. Find the analytic function whose real part $u = \dfrac{x^4 - y^4 - 2x}{x^2 + y^2}$. **04**

b. Find the bilinear transformation which maps $z = \infty, i, 0$ onto $w = -1, -i, 1$. Also find the fixed points of the transformation. **05**

c. Discuss the conformal transformation of $w = z^2$ **05**

7. a. Evaluate $\int_c \dfrac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2 (z-2)} dz$ where c is the circle (i) $|z| = 3$ (ii) $|z| = \frac{1}{2}$ (iii) $|z| = \frac{3}{2}$ by using Cauchy's integral formula. **07**

b. Expand $f(z) = \dfrac{z^2 + 1}{z^2 + 5z + 6}$ in a Laurent's series in the region (i) $2 < |Z| < 3$ (ii) $|Z| > 3$ **07**

8. a. Find the extremal of the functional $\int_a^b (y^2 + y'^2 + 2ye^x)\, dx$. **04**

b. Prove that shortest distance between two points in a plane is along the shortest line joining them. **05**

c. Prove that catenary is the curve which when rotated about a line generates a surface of minimum area. **05**